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Formulation of the multiple non-isotropic scattering process in 3-D space on the basis of energy transport theory

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SUMMARY

High-frequency seismograms of local earthquakes are considered to consist of incoherent body waves scattered by random inhomogeneities in the earth medium. For the study of seismogram envelopes, we can describe the scattering process on the basis of energy transport theory by representing the random inhomogeneities as distributed point-like scatterers. By introducing the concept of directional distribution of energy density, we propose a formulation of the multiple non-isotropic scattering process in 3-D space. We suppose that the axially symmetric non-isotropic scattering is described by a directional scattering coefficient, the medium is characterized by one wave velocity, and the source radiation is spherical. In addition to the Fourier transformation in space and the Laplace transformation in time used to solve for energy density in isotropic scattering media, we use a spherical harmonic series expansion in solid angle. Then, the energy transport equation given as an integral can be written as simultaneous linear equations, where coefficients are given by the Wigner 3-*j* symbols and the spherical harmonic expansion coefficients of the directional scattering coefficient. The lowest term of the spherical harmonic series corresponds to isotropic scattering. When the directional scattering coefficient is written as a finite-length spherical harmonic series, the simultaneous linear equations can be solved for a finite number of unknowns and we can obtain the spatio–temporal distribution of energy density. The numerical calculation for a case with strong forward scattering shows a uniform distribution of energy density around the hypocentre at large lapse times.

Key words: coda, energy transport theory, radiative transfer theory, random inhomogeneity, multiple scattering, seismogram envelope.

1 INTRODUCTION

High-frequency seismograms of local earthquakes are well characterized by their smooth envelopes, which reflect the scattering characteristics of the heterogeneous earth medium. Except for short time windows around direct *P* and *S* phases, wave trains are interpreted to be incoherent waves scattered by heterogeneities such as cracks, joints, or velocity or density anomalies that are randomly distributed in the earth. *S*-coda envelopes have been extensively studied from this point of view (Aki & Chouet 1975). It has been shown that the amplitude of the *S* coda depends on the intensity and the spatial scale of inhomogeneities, and the decay gradient is a product of a geometric factor and attenuation. One standard approach to modelling coda envelopes is to represent the heterogeneities as randomly and uniformly distributed point-like scatterers. Using this approach, the characteristics of the energy propagation have been studied based on the energy transport (or radiative transfer) theory. This method has been used for the synthesis of *S* coda not only in the single scattering approximation (Sato 1977) but also for the multiple scattering process (Hoshiba 1991; Zeng, Su & Aki 1991; Sato 1993). In most cases, isotropic scattering was assumed for mathematical simplicity; however, scattering is not isotropic, particularly at high frequencies. Non-isotropic scattering from localized inhomogeneities was explicitly investigated using the Born approximation (Sato 1984; Wu & Aki 1985). The broadening of *S*-wave envelopes was reported for intermediate-distance (100–300 km) earthquakes (Sato 1989; Scherbaum & Sato 1991). The stochastic parabolic approximation for the scalar wave

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equation was used in an attempt to explain the broadening by multiple forward scattering and diffraction due to slowly varying velocity fluctuations. In addition to diffraction effects, the importance of large-angle non-isotropic scattering due to short-wavelength fluctuations has been recently reported from observations of seismogram envelopes around the volcanic front in central Japan (Obara & Sato 1995).

The above-mentioned observations and theoretical developments lead to a need for the development of a mathematical formulation that includes multiple non-isotropic scattering processes. Frankel & Wenenberg (1987) phenomenologically introduced non-isotropic scattering in their energy-flux model. The only quantitative studies of the contribution of multiple non-isotropic scattering have used the Monte-Carlo simulation method (Abubakirov & Gusev 1990; Hoshiba 1995). The energy transport theory, which assumes the complete incoherency of scattered waves, has been thought to be very useful not only for the study of *S* coda but also for the study of total seismogram envelopes (Sato 1984); however, the theory has been used to study the contribution of non-isotropic scattering for only the single scattering process (Sato 1982), and multiple scattering has been considered only for the isotropic scattering case (Zeng *et al.* 1991). The only attempt to formulate the multiple non-isotropic scattering using the energy transport theory has been for the 2-D case of surface waves (Sato 1994b, hereafter referred to Paper I).

We propose a formulation of the multiple non-isotropic scattering process for 3-D body waves. We introduce the concept of the directional distribution of energy density, in which the integral over solid angle gives the conventional energy density. We give the integral equation for the directional distribution of energy density *a priori* in the framework of the energy transport theory. Then, we will show its relation with the conventional energy transport equation for energy density in the case of isotropic scattering, and prove the self-consistency of the formulation. In addition to the Fourier transform in space and the Laplace transform in time, we use a spherical harmonic expansion with respect to solid angle to allow us to solve the integral equation. A synthesis of the space–time distribution of energy density will be carried out for the simplest non-isotropic scattering case.

2 MATHEMATICAL FORMULATION

Here, we represent a 3-D inhomogeneous medium by a medium in which point-like scatterers are homogeneously and randomly distributed. The scattering medium has no intrinsic absorption and is characterized by one propagation velocity, V. The random distribution of scatterers makes the wavefield incoherent, so the power of scattered waves can be added. That is, we can treat the scattering process of energy density, not waves. We assume that the scattering process is non-isotropic and axially symmetric. It is characterized by a directional scattering coefficient $\zeta(\psi)$, where ψ is the scattering angle, as schematically illustrated in Fig. 1. We note that the dimension of ζ is the reciprocal of time. The total scattering coefficient is given by the integral over the solid angle Ω:

$$\zeta_T = \oint \zeta \, d\Omega = 2\pi \int_0^\pi \zeta(\psi) \sin\psi \, d\psi. \tag{1}$$

We define $f(\mathbf{r}, t; \hat{\mathbf{q}})$ as the directional distribution of energy density at $\mathbf{r}$ in direction $\hat{\mathbf{q}}$ at time t, where the circumflex denotes a unit vector. $f(\mathbf{r}, t; \hat{\mathbf{q}}) \cdot V\hat{\mathbf{q}}$ gives the energy flux density in direction $\hat{\mathbf{q}}$. The integral of f over solid angle Ω gives the energy density:

$$E(\mathbf{r}, t) = \oint f(\mathbf{r}, t; \hat{\mathbf{q}}) \, d\Omega(\hat{\mathbf{q}}). \tag{2}$$

We introduce the directional Green's function, which represents the distribution of energy density in direction $\hat{\mathbf{q}}$ due to direct propagation from a spherical impulsive radiation of unit energy located at the origin at $t = 0$ as (see Fig. 2)

$$\begin{aligned} G(\mathbf{r}, t; \hat{\mathbf{q}}) &= \frac{\exp(-\zeta_T t)}{4\pi V r^2} \delta\left(t - \frac{r}{V}\right) \delta_\Omega(\hat{\mathbf{q}}; \hat{\mathbf{r}}) \quad \text{for } t \geq 0 \\ &= 0 \qquad\qquad\qquad\qquad\qquad\quad \text{for } t < 0, \end{aligned} \tag{3}$$

where $r = |\mathbf{r}|$, $\hat{\mathbf{r}} = \mathbf{r}/r$, δ is the delta function in time, and δ_Ω is the delta function for solid angle as given by eqs (A9)–(A10). Eq. (3) means that the energy flux exists in the radial direction only. The total scattering coefficient appearing in the exponent indicates the scattering loss.

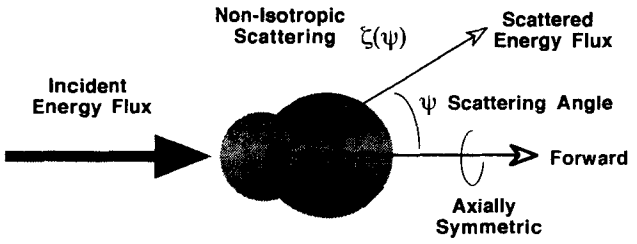


Figure 1. Axially symmetric non-isotropic scattering is characterized by the directional scattering coefficient $\zeta(\psi)$, where ψ is the scattering angle.

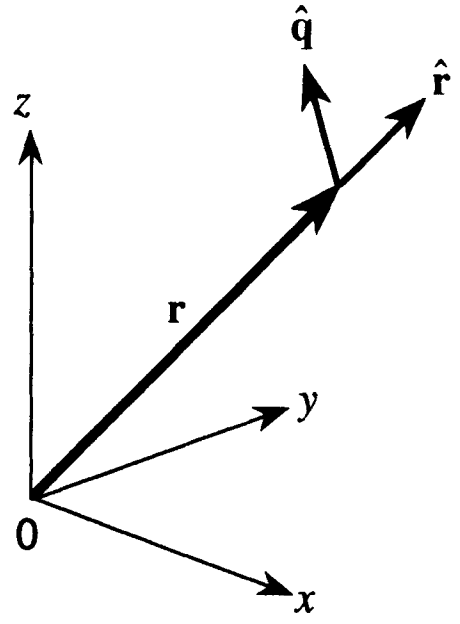


Figure 2. Geometry for the directional Green's function.

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We study the spatio–temporal change in f where the total energy W is isotropically radiated instantaneously from the origin at $t=0$. We here propose the following integral equation for describing the multiple scattering process (see Fig. 3):

$$f(\mathbf{r},t;\hat{\mathbf{q}}) = WG(\mathbf{r},t;\hat{\mathbf{q}}) + 4\pi \int_{-\infty}^{\infty}\int_{-\infty}^{\infty}\int_{-\infty}^{\infty}\int_{-\infty}^{\infty}\oint G(\mathbf{r}'',t-t';\hat{\mathbf{q}}) \times \zeta(\psi) f(\mathbf{r}',t';\hat{\boldsymbol{q}}')\, d\mathbf{r}'\, dt'\, d\Omega(\hat{\mathbf{q}}'), \quad (4)$$

where $\mathbf{r}''=\mathbf{r}-\mathbf{r}'$ is the vector from the last scatterer at $\mathbf{r}'$ to the receiver at $\mathbf{r}$. The first term in eq. (4) is the contribution due to direct propagation between the source and receiver. The second term is a convolution integral which shows that the energy flux in direction $\hat{\mathbf{q}}'$ hits the last scatterer and the scattered energy flux proportional to $\zeta(\psi)$ is produced in direction $\hat{\mathbf{r}}''$, where

$$\psi = \cos^{-1}(\hat{\mathbf{r}}''\hat{\mathbf{q}}'). \quad (5)$$

The Green's function in eq. (3) is used in the convolution integral for the propagation energy from the last scatterer to the receiver.

When scattering is isotropic, $\zeta=\zeta_T/4\pi$, by integrating eq. (4) over solid angle, we obtain

$$E_{\text{Isotropic}}(\mathbf{r},t;\hat{\mathbf{q}}) = WG_T(\mathbf{r},t) + \zeta_T \int_{-\infty}^{\infty}\int_{-\infty}^{\infty}\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} \times G_T(\mathbf{r}-\mathbf{r}',t-t')E_{\text{Isotropic}}(\mathbf{r}',t')\, d\mathbf{r}'\, dt', \quad (6)$$

where

$$G_T(\mathbf{r},t) = \frac{\exp(-\zeta_T t)}{4\pi V r^2}\,\delta\left(t-\frac{r}{V}\right) \quad \text{for } t\geq 0$$
$$= 0 \quad \text{for } t<0, \quad (7)$$

Eqs (6)–(7) show that eqs (1)–(5) are a natural extension of the conventional energy transport theory for the isotropic scattering case (Zeng *et al.* 1991).

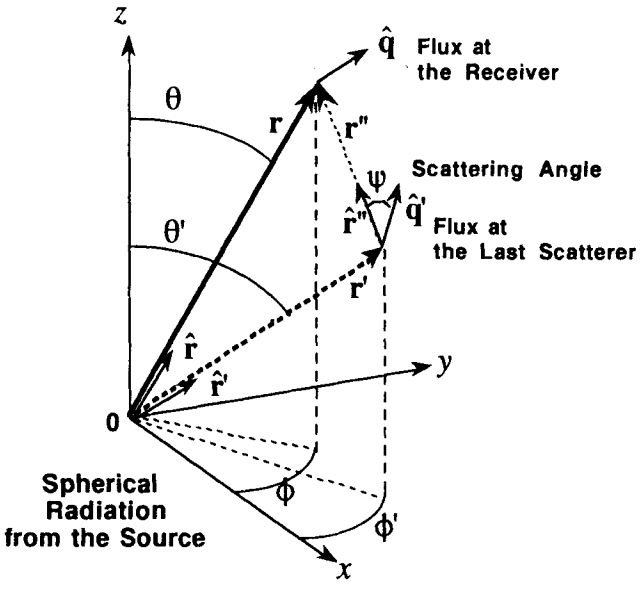


Figure 3. Geometry of the multiple non-isotropic scattering process in 3-D space.

Before solving eqs (1)–(5), we normalize time, length, and related quantities by V and ζ_T, as follows:

$$\tau=\zeta_T t, \qquad \boldsymbol{\rho}=\frac{\zeta_T}{V}\mathbf{r}, \qquad \bar{\zeta}=\frac{\zeta}{\zeta_T},$$
$$\bar{f}=\frac{f}{W(\zeta_T/V)^3}, \qquad \bar{E}=\frac{E}{W(\zeta_T/V)^3}, \quad (8)$$

where τ and $\boldsymbol{\rho}$ are the normalized time and the normalized coordinate vector, respectively. An overbar denotes the non-dimensional normalized quantity. $\bar{E}$ is the normalized energy density. Then, the integral equation (4) is rewritten as

$$\bar{f}(\boldsymbol{\rho},\tau;\hat{\mathbf{q}}) = \bar{G}(\boldsymbol{\rho},\tau;\hat{\mathbf{q}}) + 4\pi \int_{-\infty}^{\infty}\int_{-\infty}^{\infty}\int_{-\infty}^{\infty}\int_{-\infty}^{\infty}\oint \bar{G}(\boldsymbol{\rho}'',\tau-\tau';\hat{\mathbf{q}})\bar{\zeta}(\psi) \times \bar{f}(\boldsymbol{\rho}',\tau';\hat{\mathbf{q}})\, d\boldsymbol{\rho}'\, d\tau'\, d\Omega(\hat{\mathbf{q}}), \quad (9)$$

where

$$\boldsymbol{\rho}''=\boldsymbol{\rho}-\boldsymbol{\rho}', \qquad \hat{\boldsymbol{\rho}}=\hat{\mathbf{r}}, \qquad \hat{\boldsymbol{\rho}}'=\hat{\mathbf{r}}', \qquad \hat{\boldsymbol{\rho}}''=\hat{\mathbf{r}}'',$$

and (10)

$$\psi = \cos^{-1}(\hat{\boldsymbol{\rho}}''\hat{\mathbf{q}}'),$$

and the non-dimensional normalized directional Green's function is written as

$$\bar{G}(\boldsymbol{\rho},\tau;\hat{\mathbf{q}}) = \frac{e^{-\tau}}{4\pi\rho^2}\,\delta(\tau-\rho)\delta_\Omega(\hat{\mathbf{q}};\hat{\boldsymbol{\rho}}) \quad \text{for } \tau\geq 0$$
$$= 0 \quad \text{for } \tau<0, \quad (11)$$

where $\rho=|\boldsymbol{\rho}|$.

In addition to the Fourier transform in space and the Laplace transform in time, we use a spherical harmonic series expansion in solid angle instead of the Fourier series expansion that was used in the 2-D case (see Paper I) to solve the integral equation (9). We define the Fourier transform in 3-D space as

$${}^{F}\bar{f}(\mathbf{k},\tau) = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} d\boldsymbol{\rho}\, e^{-i\mathbf{k}\cdot\boldsymbol{\rho}}\bar{f}(\boldsymbol{\rho},\tau;\hat{\mathbf{q}}). \quad (12\text{-}1)$$

The Laplace transform in time is

$${}^{L}\tilde{f}(\boldsymbol{\rho},s) = \int_0^{\infty} d\tau\, e^{-s\tau}\bar{f}(\boldsymbol{\rho},\tau;\hat{\mathbf{q}}). \quad (12\text{-}2)$$

Finally, we define the spherical harmonic expansion in solid angle as

$${}^{Y}\bar{f}_{lm}(\boldsymbol{\rho},\tau) = \oint d\Omega(\hat{\mathbf{q}})Y^*_{lm}(\theta_q,\phi_q)\bar{f}(\boldsymbol{\rho},\tau;\hat{\mathbf{q}}), \quad (12\text{-}3)$$

where Y_{lm} is the spherical harmonic function as given by (A1)–(A5) and $\hat{\mathbf{q}}=(1,\theta_q,\phi_q)$ in spherical coordinates. We adopt the phase convention after Landau & Lifshitz (1989). The Laplace transform in time and the spherical harmonic expansion in solid angle is

$${}^{LY}\bar{f}_{lm}(\boldsymbol{\rho},s) = \int_0^{\infty} d\tau\, e^{-s\tau}\oint d\Omega(\hat{\mathbf{q}})Y^*_{lm}(\theta_q,\phi_q)\bar{f}(\boldsymbol{\rho},\tau;\hat{\mathbf{q}}), \quad (12\text{-}4)$$

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The Fourier transform in space, the Laplace transform in time, and the spherical harmonic expansion in solid angle is

$$^{\mathrm{FLY}}\bar{f}_{lm}(\mathbf{k},s)=\int_{-\infty}^{\infty}\int_{-\infty}^{\infty}\int_{-\infty}^{\infty}d\boldsymbol{\rho}\mathrm{e}^{-i\mathbf{k}\cdot\boldsymbol{\rho}}\int_{0}^{\infty}d\tau\mathrm{e}^{-s\tau}\times\oint d\Omega(\hat{\mathbf{q}})Y_{lm}^{*}(\theta_q,\phi_q)\bar{f}(\boldsymbol{\rho},\tau;\hat{\mathbf{q}}). \tag{12-5}$$

The inverse transform is

$$\bar{f}(\boldsymbol{\rho},\tau;\hat{\mathbf{q}})=\frac{1}{(2\pi)^3}\int_{-\infty}^{\infty}\int_{-\infty}^{\infty}\int_{-\infty}^{\infty}d\mathbf{k}\mathrm{e}^{i\mathbf{k}\cdot\boldsymbol{\rho}}\frac{1}{2\pi i}\int_{-\infty i}^{\infty i}ds\mathrm{e}^{\tau s}\times\sum_{l=0}^{\infty}\sum_{m=-l}^{l}Y_{lm}(\theta_q,\phi_q)^{\mathrm{FLY}}\bar{f}_{lm}(\mathbf{k},s). \tag{13}$$

Taking the Laplace transformation with respect to time and the spherical harmonic series expansion of (11) by using (A9)–(A10), we obtain

$$^{\mathrm{LY}}\bar{G}_{lm}(\boldsymbol{\rho},s)=\frac{\mathrm{e}^{-(s+1)\rho}}{4\pi\rho^2}Y_{lm}^{*}(\theta,\phi), \tag{14}$$

where subscripts and superscripts have the same meaning as when used with $\bar{f}$. Taking the Fourier transform of (14) with respect to space coordinates, we obtain the product of a spherical harmonic function and a function of $k=|\mathbf{k}|$, $\bar{\bar{G}}_l$, as

$$^{\mathrm{FLY}}\bar{G}_{lm}(\mathbf{k},s)=\int_{-\infty}^{\infty}\int_{-\infty}^{\infty}\int_{-\infty}^{\infty}d\boldsymbol{\rho}\mathrm{e}^{-i\mathbf{k}\cdot\boldsymbol{\rho}}{}^{\mathrm{LY}}\bar{G}_{lm}(\boldsymbol{\rho},s)=(-i)^l\bar{\bar{G}}_l(k,s)Y_{lm}^{*}(\theta_k,\phi_k), \tag{15}$$

where $\mathbf{k}=(k,\theta_k,\phi_k)$ in spherical coordinates. It is explicitly written using Gauss's hypergeometric function (see Gradshteyn & Ryzhik 1980, p. 711) as

$$\bar{\bar{G}}_l(k,s)=\int_0^{\infty}j_l(k\rho)\mathrm{e}^{-(s+1)\rho}\,d\boldsymbol{\rho}=\frac{1}{k}\left(\frac{k}{2(s+1)}\right)^{l+1}\frac{\sqrt{\pi}\,\Gamma(l+1)}{\Gamma(l+\frac{3}{2})}\times{}_2F_1\left[\frac{l+1}{2},\frac{l+2}{2},l+\frac{3}{2},-\left(\frac{k}{s+1}\right)^2\right]. \tag{16}$$

Next, we expand the non-dimensional normalized directional scattering coefficient using spherical harmonic functions as

$$\bar{\zeta}(\psi)=\sum_{l=0}^{\infty}(-i)^l\bar{\zeta}_lY_{l,0}(\cos^{-1}(\hat{\boldsymbol{\rho}}''\hat{\mathbf{q}}'),0)=\sum_{l=0}^{\infty}\sqrt{\frac{4\pi}{2l+1}}\bar{\zeta}_l\sum_{m=-l}^{l}Y_{lm}(\theta'',\phi'')Y_{lm}^{*}(\theta_{q'},\phi_{q'}), \tag{17}$$

$$\bar{\zeta}(\psi)=\bar{\zeta}_0Y_{00}(\psi,0)-i\bar{\zeta}_1Y_{10}(\psi,0)-\bar{\zeta}_2Y_{20}(\psi,0)+\cdots$$



Figure 4. Schematic illustration of the spherical harmonic series expansion of the normalized directional scattering coefficient.

where we used the addition theorem as given by (A6). The expansion coefficients $\bar{\zeta}_l$'s are real quantities and $\bar{\zeta}_0=1/\sqrt{4\pi}$ from (1) and (8). We note that the lowest term with $l=0$ corresponds to isotropic scattering. This expansion is schematically illustrated in Fig. 4. Substituting (17) into (9), we obtain

$$\bar{f}(\boldsymbol{\rho},s;\hat{\mathbf{q}})=\bar{G}(\boldsymbol{\rho},s;\hat{\mathbf{q}})+4\pi\int_{-\infty}^{\infty}\int_{-\infty}^{\infty}\int_{-\infty}^{\infty}\bar{G}(\boldsymbol{\rho}'',s;\hat{\mathbf{q}})\times\sum_{l'=0}^{\infty}\sqrt{\frac{4\pi}{2l'+1}}\bar{\zeta}_{l'}\sum_{m'=-l'}^{l'}Y_{l'm'}(\theta'',\phi'')\times{}^{\mathrm{LY}}\bar{f}_{l'm'}(\boldsymbol{\rho}',s)\,d\boldsymbol{\rho}'. \tag{18}$$

Substituting (12-4) into (18) and using orthogonal relation, (A5), we have

$$^{\mathrm{LY}}\bar{f}_{lm}(\boldsymbol{\rho},s)={}^{\mathrm{LY}}\bar{G}_{lm}(\boldsymbol{\rho},s)+4\pi\sum_{l'=0}^{\infty}\sqrt{\frac{4\pi}{2l'+1}}\bar{\zeta}_{l'}\sum_{m'=-l'}^{l'}\times\int_{-\infty}^{\infty}\int_{-\infty}^{\infty}\int_{-\infty}^{\infty}{}^{\mathrm{LY}}\bar{G}_{lm}(\boldsymbol{\rho}'',s)Y_{l'm'}(\theta'',\phi'')\times{}^{\mathrm{LY}}\bar{f}_{l'm'}(\boldsymbol{\rho}',s)\,d\boldsymbol{\rho}'. \tag{19}$$

By using (13), (14) and (A7), we obtain

$$\int_{-\infty}^{\infty}\int_{-\infty}^{\infty}\int_{-\infty}^{\infty}{}^{\mathrm{LY}}\bar{G}_{lm}(\boldsymbol{\rho},s)Y_{l'm'}(\theta,\phi)\mathrm{e}^{-i\mathbf{k}\cdot\boldsymbol{\rho}}\,d\boldsymbol{\rho}=\sum_{l''=0}^{\infty}\sum_{m''=-l''}^{l''}(Y_{l''m''})_{l'm'}^{lm}{}^{\mathrm{FLY}}\bar{G}_{l''m''}(\mathbf{k},s), \tag{20}$$

where $(Y_{l''m''})_{l'm'}^{lm}$ on the right-hand side is explicitly written by using the Wigner 3-j symbols (see A11). Taking the Fourier transform of (19) with respect to space coordinates and substituting (20) into the result, we finally obtain

$$^{\mathrm{FLY}}\bar{f}_{lm}(\mathbf{k},s)={}^{\mathrm{FLY}}\bar{G}_{lm}(\mathbf{k},s)+4\pi\sum_{l'=0}^{\infty}\sqrt{\frac{4\pi}{2l'+1}}\bar{\zeta}_{l'}\times\sum_{m'=-l'}^{l'}\sum_{l''=0}^{\infty}\sum_{m''=-l''}^{l''}(Y_{l''m''})_{l'm'}^{lm}\times{}^{\mathrm{FLY}}\bar{G}_{l''m''}(\mathbf{k},s)^{\mathrm{FLY}}\bar{f}_{l'm'}(\mathbf{k},s). \tag{21}$$

These simultaneous linear equations for $0\le l<\infty$ and $-l\le m\le l$ are the energy transport equation in the conjugate space. We may factorize $^{\mathrm{FLY}}\bar{f}_{lm}$ into a spherical harmonic function and a function of k as in (15):

$$^{\mathrm{FLY}}\bar{f}_{lm}(\mathbf{k},s)=(-i)^l\bar{\bar{f}}_l(k,s)Y_{lm}^{*}(\theta_k,\phi_k). \tag{22}$$

This implicitly means that $\bar{f}(\boldsymbol{\rho},\tau;\hat{\mathbf{q}})$ is a function of $\psi=\cos^{-1}(\hat{\boldsymbol{\rho}}\hat{\mathbf{q}})$, since there is no specific direction in this system. Substituting (22) into (21) and integrating over solid angle, we obtain the master equation for $\bar{\bar{f}}_l$ for $l\ge0$ as

$$\bar{\bar{f}}_l(k,s)=\bar{\bar{G}}_l(k,s)+4\pi\sum_{l'=0}^{\infty}\sqrt{\frac{4\pi}{2l'+1}}\bar{\zeta}_{l'}\sum_{m'=-l'}^{l'}\sum_{l''=0}^{\infty}\times(-i)^{l'+l''-l}\sum_{m''=-l''}^{l''}|(Y_{l''m''})_{l'm'}^{lm'+m''}|^2\times\bar{\bar{G}}_{l''}(k,s)\bar{\bar{f}}_{l'}(k,s). \tag{23}$$

We note that $\bar{\bar{f}}_l$ and $\bar{\bar{G}}_l$ are real functions and the power of $-i$, $l'+l''-l$, is even for non vanishing $(Y_{l''m''})_{l'm'}^{lm'+m''}$.

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It must be emphasized that, although the mathematical derivation involves spherical expansion of $\bar{\bar{f}}_l$ for $l=0$ to infinity, the actual computation of the energy density requires $l=0$ only. If the directional scattering coefficient is written as a finite number of terms of the spherical harmonic series (17), the triangle inequality for the Wigner 3-j symbols makes the right-hand side of (23) a finite series for a given l. That is, if the highest order of $\bar{\zeta}_l$ is $l_{\max}$, we need to solve the simultaneous linear equations for $l_{\max}+1$ unknowns. Then, we can formally obtain the normalized energy density $\bar{E}(\boldsymbol{\rho}, \tau)$ from $\bar{\bar{f}}_0(k, s)$:

$$\bar{E}(\boldsymbol{\rho}, \tau) = \oint \bar{f}(\boldsymbol{\rho}, \tau; \hat{\mathbf{q}})\, d\Omega(\hat{\mathbf{q}})$$

$$= \frac{1}{\rho(2\pi)^2} \int_{-\infty}^{\infty} d\omega e^{-i\omega\tau} \int_{-\infty}^{\infty} dk e^{-ik\rho} \times \left[\frac{ik}{2\pi} \bar{\bar{f}}_0(k \to -k, s \to -i\omega)\right]. \quad (24)$$

We can numerically evaluate this integral by using a 2-D Fast Fourier transform (FFT).

3 ENERGY CONSERVATION

Taking the limit $k \to 0$ of (16), we obtain

$$\bar{\bar{G}}_n(0, s) = \frac{1}{s+1} \delta_{n0}. \quad (25)$$

Substituting (25) into (23) and using $\bar{\zeta}_0 = 1/\sqrt{4\pi}$, we have

$$\int_0^{\infty} d\tau e^{-s\tau} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} d\boldsymbol{\rho} \bar{E}(\boldsymbol{\rho}, \tau) = \bar{\bar{f}}_0(0, s) = \frac{1}{s}. \quad (26)$$

This means that the total energy given by the space integral of $\bar{E}$ is the same at any time:

$$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \bar{E}(\boldsymbol{\rho}, \tau)\, d\boldsymbol{\rho} = 1. \quad (27)$$

The conservation of total energy ensures the self consistency of the set of equations (1)–(5) for describing the multiple non-isotropic scattering process.

4 PRACTICAL SIMULATION METHOD

Following the procedure used in Paper I, we show a practical simulation method. We first decompose each of $\bar{\bar{f}}_0$ and $\bar{E}$ into three terms corresponding to the direct, the single scattering and the multiple scattering with order greater than or equal to two:

$$\bar{\bar{f}}_0(k, s) = \bar{\bar{G}}_0(k, s) + \bar{\bar{f}}_0^1(k, s) + \bar{\bar{f}}_0^{\mathrm{M}}(k, s), \quad (28)$$

$$\bar{E}(\boldsymbol{\rho}, \tau) = \bar{G}(\boldsymbol{\rho}, \tau) + \bar{E}^1(\boldsymbol{\rho}, \tau) + \bar{E}^{\mathrm{M}}(\boldsymbol{\rho}, \tau). \quad (29)$$

We know that the convergence of integral (24) over k and ω is very slow because of the direct and the single scattering terms: the former is a delta function and the latter has a logarithmic divergence at the direct-wave front in the time domain, even for the isotropic scattering case (Sato 1977). There is a technique to calculate numerically such a slow convergent integral (Zeng 1993); however, we will use integral (24) only for the multiple scattering term, which converges quickly. Substituting $\bar{\bar{G}}_{l'}$ for $\bar{\bar{f}}_{l'}$ on the right-hand side of (23), we get the single scattering term:

$$\bar{\bar{f}}_0^1(k, s) = 4\pi \sum_{l'=0}^{\infty} \sqrt{\frac{4\pi}{2l'+1}}\, \bar{\zeta}_{l'} \sum_{m'=-l'}^{l'} \sum_{l''=0}^{\infty} (-i)^{l'+l''} \sum_{m''=-l''}^{l''} \times |(Y_{l',m'})_{l'',m''}^{0,m'+m''}|^2\, \bar{\bar{G}}_{l'}(k, s) \bar{\bar{G}}_{l''}(k, s). \quad (30)$$

Substituting $\bar{\bar{f}}_0^1$ and $\bar{\bar{G}}_0$ into (28), where $\bar{\bar{f}}_0$ is the solution of (23), we get $\bar{\bar{f}}_0^{\mathrm{M}}$. By using a FFT we can numerically calculate $\bar{E}^{\mathrm{M}}(\boldsymbol{\rho}, \tau)$.

For the calculation of the single scattering term $\bar{E}^1(\boldsymbol{\rho}, \tau)$, we do not use (30) but directly integrate the single non-isotropic scattering in space (see Sato 1982). Substituting $\bar{G}_0$ into $\bar{f}$ in the second term of the right-hand side of (9), we obtain the single scattering term $\bar{f}^1(\boldsymbol{\rho}, \tau; \hat{\mathbf{q}})$, of which the integral over solid angle gives $\bar{E}^1(\boldsymbol{\rho}, \tau)$. We take the source at the origin and the receiver on the z'-axis as the foci of the prolate spheroidal coordinates (v, u, ϕ) (see Fig. 5):

$$x' = \frac{\rho}{2}\sqrt{v^2-1} \sin u \cos\phi,$$

$$y' = \frac{\rho}{2}\sqrt{v^2-1} \sin u \sin\phi, \quad (31)$$

$$z' = \frac{\rho}{2}(1 + v \cos u),$$

where $1 \le v < \infty$, $0 \le u \le \pi$, and $0 \le \phi < 2\pi$. Then,

$$\bar{E}^1(\boldsymbol{\rho}, \tau) = \oint \bar{f}^1(\boldsymbol{\rho}, \tau; \hat{\mathbf{q}})\, d\Omega(\hat{\mathbf{q}})$$

$$= 4\pi \oint \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \oint \bar{G}(\boldsymbol{\rho}'', \tau - \tau'; \hat{\mathbf{q}}) \bar{\zeta}(\psi) \times \bar{G}(\boldsymbol{\rho}', \tau'; \hat{\mathbf{q}}')\, d\boldsymbol{\rho}'\, d\tau'\, d\Omega(\hat{\mathbf{q}}')\, d\Omega(\hat{\mathbf{q}})$$

$$= 4\pi \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \oint \oint \times \frac{e^{-(\tau-\tau')} \delta(\tau - \tau' - \rho'') \delta_{\Omega}(\hat{\mathbf{q}}; \hat{\boldsymbol{\rho}}'') \bar{\zeta}[\cos^{-1}(\hat{\boldsymbol{\rho}}'' \hat{\mathbf{q}}')]}{4\pi\rho''^2} \times \frac{e^{-\tau'} \delta(\tau' - \rho') \delta_{\Omega}(\hat{\mathbf{q}}'; \hat{\boldsymbol{\rho}})}{4\pi\rho'^2} d\boldsymbol{\rho}'\, d\tau'\, d\Omega(\hat{\mathbf{q}}')\, d\Omega(\hat{\mathbf{q}})$$

$$= \frac{e^{-\tau}}{4\pi} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \frac{\delta(\tau - \rho' - \rho'') \bar{\zeta}[\cos^{-1}(\hat{\boldsymbol{\rho}}'' \hat{\boldsymbol{\rho}}')]}{\rho''^2 \rho'^2} d\boldsymbol{\rho}'$$

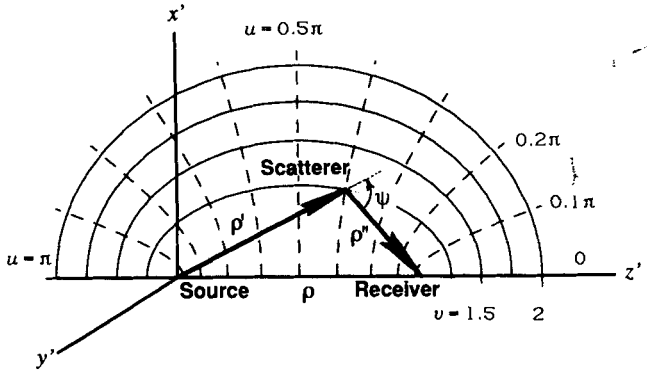


Figure 5. Geometry of the single non-isotropic scattering process. The source and the receiver, located on the z' axis, are taken as the foci of the prolate spheroidal coordinate system.

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$$= \frac{e^{-\tau}}{4\pi} \int_1^\infty dv \int_0^\pi \sin u \, du \int_0^{2\pi} d\phi \times \frac{\delta(\tau - \rho' - \rho'')\bar{\zeta}[\psi(u, v)]}{\rho''^2 \rho'^2} \frac{\rho \rho' \rho''}{2}$$

$$= \frac{e^{-\tau}}{\rho^2} H\left(\frac{\tau}{\rho} - 1\right) \int_0^\pi \frac{\bar{\zeta}\left[\psi\left(u, \frac{\tau}{\rho}\right)\right] \sin u}{(\tau/\rho)^2 - \cos^2 u} du, \quad (32)$$

where H is the step function. The single scattering term given by the surface integral over the scattering shell corresponding to $v = \tau/\rho$ is written as an integral over the prolate spheroidal coordinate u. The scattering angle ψ is explicitly written as (see Sato 1984)

$$\psi(u, v) = \cos^{-1}(\hat{\boldsymbol{\rho}}'' \hat{\boldsymbol{\rho}}') = \cos^{-1}\left(\frac{2 - v^2 - \cos^2 u}{v^2 - \cos^2 u}\right). \quad (33)$$

In the case of isotropic scattering we can analytically integrate (32) (Sato 1977); however, we have to integrate (32) numerically in general.

Thus, by adding the direct term $\bar{G}_0(\boldsymbol{\rho}, \tau)$, the single scattering term $\bar{E}^1(\boldsymbol{\rho}, \tau)$ given by a numerical integral in space, and the multiple scattering term $\bar{E}^M(\boldsymbol{\rho}, \tau)$ given by the two-dimensional FFT, we get the spatio-temporal distribution of the normalized energy density $\bar{E}(\boldsymbol{\rho}, \tau)$.

5 EXAMPLE

We examine the mathematical structure of our formulation. We take the following simple non-isotropic scattering with one parameter $\bar{\zeta}_1 = 0.08$ as an example. Forward scattering is stronger than backward scattering:

$$\bar{\zeta}(\psi) = \bar{\zeta}_0 Y_{00}(\psi, 0) - i\bar{\zeta}_1 Y_{10}(\psi, 0) = \frac{1}{4\pi} + 0.08\sqrt{\frac{3}{4\pi}} \cos\psi. \quad (34)$$

We would like to show how to calculate the spatio-temporal distribution of the normalized energy density. The scattering pattern is shown in Fig. 6, where isotropic scattering is plotted with a broken curve for reference. By using the explicit values of the Wigner 3-j symbols as given by (A12), we write the first two linear equations of (23) as

$$\bar{\bar{f}}_0 = \bar{\bar{G}}_0 + \bar{\bar{G}}_0 \bar{\bar{f}}_0 - 2\sqrt{3\pi}\,\bar{\zeta}_1 \bar{\bar{G}}_1 \bar{\bar{f}}_1$$
$$\bar{\bar{f}}_1 = \bar{\bar{G}}_1 + 3\bar{\bar{G}}_1 \bar{\bar{f}}_0 + 2\sqrt{3\pi}\,\bar{\zeta}_1 \bar{\bar{G}}_0 \bar{\bar{f}}_1 - 4\sqrt{3\pi}\,\bar{\zeta}_1 \bar{\bar{G}}_2 \bar{\bar{f}}_1 \quad (35)$$

Substituting $\bar{\bar{G}}_l$ into $\bar{\bar{f}}_l$ in the right-hand side of the first equation, we find that the single scattering term is

$$\bar{\bar{f}}_0^1 = \bar{\bar{G}}_0 + \bar{\bar{G}}_0 \bar{\bar{G}}_0 - 2\sqrt{3\pi}\,\bar{\zeta}_1 \bar{\bar{G}}_1 \bar{\bar{G}}_1. \quad (36)$$

Subtracting $\bar{\bar{G}}_0$ and $\bar{\bar{f}}_0^1$ from the solution $\bar{\bar{f}}_0$ of (35), we obtain $\bar{\bar{f}}_0^M$. Then, we obtain $\bar{E}^M(\boldsymbol{\rho}, \tau)$ by using (24), where

$$\bar{\bar{G}}_0(k, s) = \frac{1}{k} \tan^{-1} \frac{k}{s+1},$$
$$\bar{\bar{G}}_1(k, s) = \frac{1}{k}\left(1 - \frac{s+1}{k} \tan^{-1} \frac{k}{s+1}\right), \quad (37)$$
$$\bar{\bar{G}}_2(k, s) = \frac{1}{2k}\left\{-3\left(\frac{s+1}{k}\right) + \left[3\left(\frac{s+1}{k}\right)^2 + 1\right] \tan^{-1} \frac{k}{s+1}\right\}.$$

We plot the resultant temporal traces and the spatial sections of the normalized energy density in Fig. 7. The energy density corresponding to isotropic scattering is plotted by a broken curve for reference in each plot. On the left of Fig. 7, we find that the temporal traces are very close to the isotropic scattering case for short distances. Strong forward scattering increases the energy density just after the direct arrival, and weak backward scattering decreases the latter coda energy density compared with the isotropic scattering case as lapse time increases. Non-isotropic scattering apparently has a strong effect, even at times when multiple scattering dominates the envelope. At larger distances, a secondary peak appears just following the direct arrival, and is much earlier than in the isotropic scattering case. If forward scattering is stronger than that given by eq. (34), the secondary peak will appear closer to the direct arrival. Thus, we may say that both the direct and the secondary peaks constitute the broadened envelope due to multiple forward scattering, which has previously been studied on the basis of strong diffraction and multiple path effects (Sato 1989). On the right portion of Fig. 7, we find that the difference from the isotropic scattering case is apparent, even in the top spatial section, at short lapse time. At large lapse times, the spatial section is rather uniform in space compared with the isotropic scattering case, which has a Gaussian-like peak. We find that the secondary peak grows due to multiple forward scattering in the second and the third spatial sections. However, the secondary peak diminishes with increasing lapse time, as shown in the bottom spatial section. If forward scattering is much stronger than that given by (34), the secondary peak near the direct propagation front will remain in the spatial section for longer.

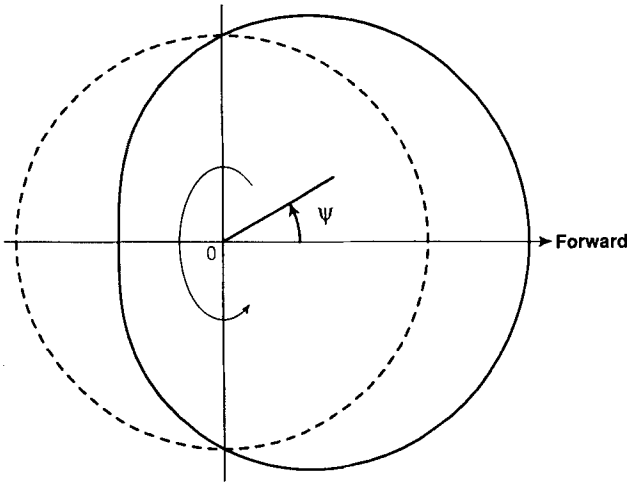


Figure 6. Angular dependence of the normalized directional scattering coefficient (34) (solid curve), the broken curve shows the isotropic scattering for reference.

6 SUMMARY AND DISCUSSION

By the introduction of concepts of the directional distribution of energy density and the directional Green's function, we have formulated the multiple non-isotropic scattering process in 3-D on the basis of the energy

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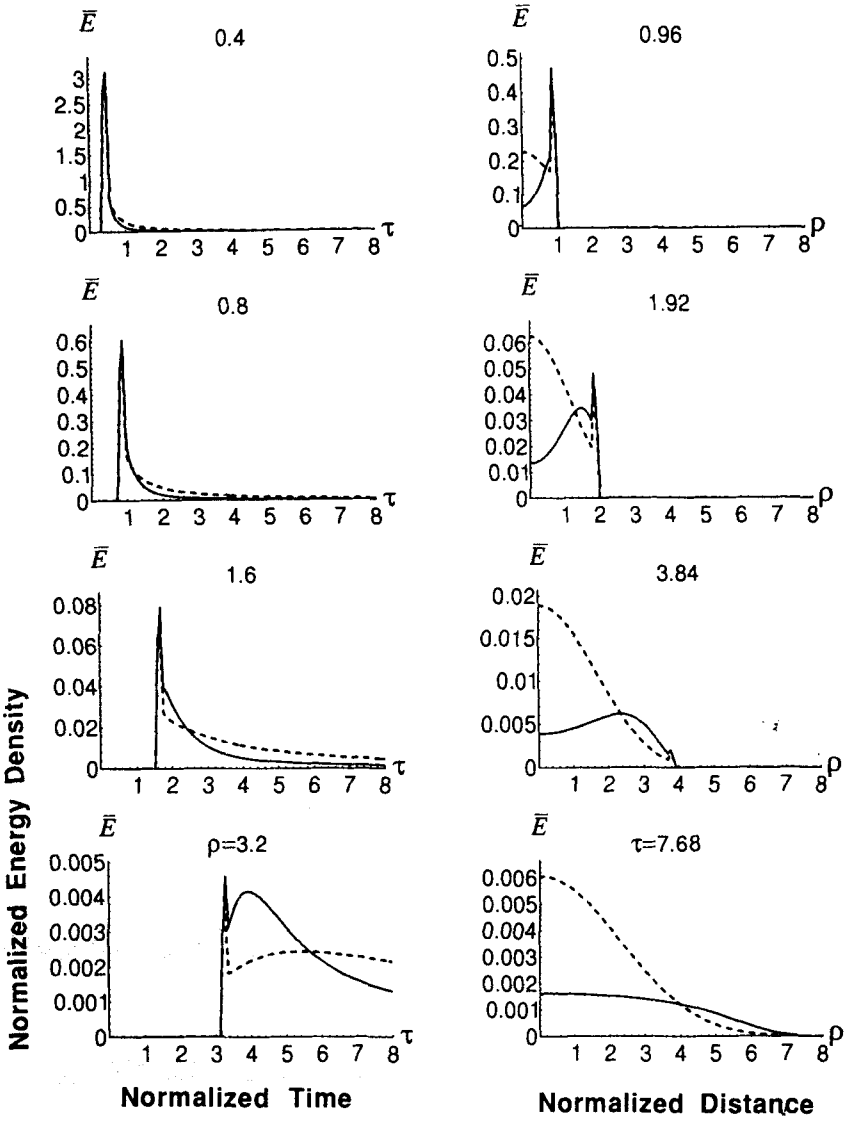


Figure 7. Temporal traces (left) and spatial sections (right) of the normalized energy density for the non-isotropic scattering (34), where a broken curve corresponds to the isotropic scattering for reference. A two-dimensional FFT is done over 200×200 points for $((0–16), (0–16))$ in the normalized space-time (ρ, τ). The source duration time is taken to be 0.16, and has two samples. The number in each plot is the normalized distance (left) or the normalized lapse time (right).

transport theory. The total energy conservation ensures the mathematical self-consistency of the formulation. By using the Fourier transform in space, the Laplace transform in time, and the spherical harmonic series expansion with respect to solid angle, the energy transport equation is written as simultaneous linear equations, in which the spherical harmonic series expansion coefficients of the directional scattering coefficient and Wigner 3-*j* symbols appear. When the non-isotropic scattering is described by a finite number of terms of the spherical harmonic series, we can obtain the energy density by solving a set of linear equations for a finite number of unknowns. In the case of strong forward scattering, a concentration of energy is found just after the direct arrival and less coda energy is excited at large lapse times, as compared with the isotropic scattering case. The secondary peak appears much closer to the direct arrival in the time trace with increasing distance compared with the isotropic scattering case. The secondary peak appears near the propagation front in the spatial section and diminishes with increasing lapse time. The energy density is found to be spatially nearly uniform around the source, even at large lapse times compared with the isotropic scattering case. This result was predicted from a similar study in 2-D (Paper I). The coda normalization method for attenuation measurements and site factor estimation (Aki 1980) is based on the hypothesis that the coda energy is uniformly distributed. Our simple case study suggests a possible way to explain this.

For the study of stronger forward scattering cases, we will have to include additional higher order terms of the spherical harmonic series with respect to solid angle. We will also have to obtain the Green's functions in the conjugate space for higher orders in an analytical form such as (37) for numerical calculation. Then, we will be able to simulate seismogram envelopes for more realistic non-isotropic scattering amplitudes of body waves derived from the wave

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equation approach. Our formulation will make it possible to study the contribution of wide-angle scattering to the broadening of seismogram envelopes, which has previously only been studied using the parabolic approximation. For the study of total seismogram envelopes, it will also be necessary to formulate the multiple non-isotropic scattering process with conversions between P and S. Scattering amplitudes of P–S converted waves have been shown to have several lobes, but the formulation has been done only for the isotropic scattering case (Sato 1994a).

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APPENDIX A

We here follow the definition of the spherical harmonic function in 'Quantum Mechanics' (Landau & Lifschitz 1989). We first define

$$\Theta_{lm}(\theta) = (-1)^m i^l \sqrt{\frac{(2l+1)}{2}\frac{(l-m)!}{(l+m)!}} P_l^m(\cos\theta) \quad \text{for } m \geq 0,$$

$$= i^l \sqrt{\frac{(2l+1)}{2}\frac{(l-|m|)!}{(l+|m|)!}} P_l^{|m|}(\cos\theta) \quad \text{for } m < 0, \tag{A1}$$

where P_l^m are associated Legendre polynomials and the choice of the phase factor i^l is different from the usual definition as in Abramowitz & Stegun (1970). This choice, however, is the most natural from the point of view of the theory of addition of angular momenta. Then

$$\Phi_m(\phi) = \frac{1}{\sqrt{2\pi}} e^{im\phi}. \tag{A2}$$

The normalized spherical harmonic function is given by

$$Y_{lm}(\theta, \phi) = \Theta_{lm}(\theta)\Phi_m(\phi). \tag{A3}$$

We note that

$$Y^*_{l,m}(\theta, \phi) = (-1)^{l-m} Y_{l,-m}(\theta, \phi). \tag{A4}$$

The normalized spherical harmonic functions with different l or m are orthogonal:

$$\oint Y^*_{l_1 m_1}(\theta, \phi) Y_{l_2 m_2}(\theta, \phi) \sin\theta \, d\theta \, d\phi = \delta_{l_1 l_2}\delta_{m_1 m_2}. \tag{A5}$$

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The following 'addition theorem' holds for Legendre polynomials:

$$P_l[\cos\theta\cos\theta' + \sin\theta\sin\theta'\cos(\phi-\phi')]$$

$$= \frac{4\pi}{2l+1}\sum_{m=-l}^{l} Y_{lm}(\theta,\phi)Y^*_{lm}(\theta',\phi')$$

$$= \frac{4\pi}{2l+1}\sum_{m=-l}^{l} Y^*_{lm}(\theta,\phi)Y_{lm}(\theta',\phi'). \quad \text{(A6)}$$

We note the following expansion:

$$\exp\{ikr[\cos\theta\cos\theta' + \sin\theta\sin\theta'\cos(\phi-\phi')]\}$$

$$= 4\pi\sum_{l=0}^{\infty} i^l j_l(kr)\sum_{m=-l}^{l} Y_{lm}(\theta,\phi)Y^*_{lm}(\theta',\phi'), \quad \text{(A7)}$$

where the spherical Bessel function is

$$j_l(z) = \sqrt{\frac{\pi}{2z}}J_{l+1/2}(z). \quad \text{(A8)}$$

From (A6), we define the delta function for solid angle (see Terasawa 1960, p. 55) as

$$\delta_\Omega(\hat{\mathbf{r}};\hat{\mathbf{r}}') = \delta_\Omega(\theta,\phi;\theta',\phi')$$

$$= \sum_{l=0}^{\infty}\sum_{m=-l}^{l} Y_{lm}(\theta,\phi)Y^*_{lm}(\theta',\phi')$$

$$= \sum_{l=0}^{\infty}\sum_{m=-l}^{l} Y^*_{lm}(\theta,\phi)Y_{lm}(\theta',\phi'), \quad \text{(A9)}$$

where $\hat{\mathbf{r}} = (1,\theta,\phi)$ and $\hat{\mathbf{r}}' = (1,\theta',\phi')$ in spherical coordinates. We note that

$$\oint\delta_\Omega(\theta,\phi;\theta',\phi')\,d\Omega(\theta',\phi') = 1$$

and (A10)

$$\oint f(\theta',\phi')\delta_\Omega(\theta,\phi;\theta',\phi')\,d\Omega(\theta',\phi') = f(\theta,\phi) \quad \text{for any } f$$

The integral of a product of three spherical harmonic functions is written by using the Wigner 3-j symbols (see Landau & Lifshitz 1989):

$$(Y_{lm})^{l_1 m_1}_{l_2 m_2} \equiv \oint Y^*_{l_1 m_1}(\theta,\phi)Y_{lm}(\theta,\phi)Y_{l_2 m_2}(\theta,\phi)\,d\Omega$$

$$= (-1)^{m_1} i^{l-l_1+l_2}\begin{pmatrix} l_1 & l & l_2 \\ -m_1 & m & m_2\end{pmatrix}$$

$$\times\begin{pmatrix} l_1 & l & l_2 \\ 0 & 0 & 0\end{pmatrix}\sqrt{\frac{(2l_1+1)(2l+1)(2l_2+1)}{4\pi}}. \quad \text{(A11)}$$

There are selection rules corresponding to the addition of angular momenta. The above integral vanishes except when $m_1 = m + m_2$ and $|l-l_2| \le l_1 \le |l+l_2|$ (the triangular condition). In this case, $l + l_1 + l_2$ is even. We note that the Wigner 3-j symbols are pure real. The square of (A11) is used in the text:

$$|(Y_{lm})^{l_1 m_1}_{l_2 m_2}|^2 = \frac{1}{4\pi}\begin{pmatrix} l_1 & l & l_2 \\ -m_1 & m & m_2\end{pmatrix}^2\begin{pmatrix} l_1 & l & l_2 \\ 0 & 0 & 0\end{pmatrix}^2$$

$$\times(2l_1+1)(2l+1)(2l_2+1). \quad \text{(A12)}$$

Shriner & Thompson (1993) recently reviewed the published programs for the numerical computation of the Wigner 3-j symbols as done in Mathematica (Wolfram 1991).

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